Hyunwoo (Shawn) Kim

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Research Interests

Classical simulation of quantum circuits and the quantum–classical boundary (Pauli propagation, classical surrogates), quantum machine learning and quantum generative models, HPC–QPU hybrid workflows, hardware-aware NISQ algorithms and quantum error mitigation

Education

Sogang University — Seoul, Korea
B.S. in Physics & Artificial Intelligence (double major) — Mar 2020 – Feb 2026

- GPA 3.81/4.30 (95.8th percentile), *Magna Cum Laude*; four-year full-tuition merit scholarship
- Coursework: Linear Algebra, Mathematical Physics I–II, Quantum Physics I–II, Machine Learning, Deep Learning
- Credit exchange at Yonsei: Quantum Computing (fall 2024), Quantum Information Science (spring 2025)

Research Experience

NORMA, Inc., Quantum AI Team — Seoul, Korea
Research Scientist — Jul 2025 – Present

GPU-accelerated quantum simulation

- Co-developed PADO-Pauli, an open-source, differentiable Pauli-propagation engine in PyTorch: a circuit is compiled once into sparse tensor maps, then evaluated and back-propagated in batch on CUDA and ROCm GPUs
- Zero filtering removes propagated Pauli terms that cannot contribute to the result, giving about 1.5× faster batched evaluation and 1.4–3.0× lower steady-state memory
- Against NVIDIA cuPauliProp 0.4.0 on one A100 80 GB (depth-3 grid-Ising QAOA): 2.7–10.1× faster evaluation, 3.3–13.3× faster forward and backward passes, and 206× higher throughput on batched inputs
- Used 3.1× less GPU memory at 16 qubits (14.5 vs. 45.4 GB) and completed a setting in which cuPauliProp ran out of the 80 GB device memory
- Reproduced IBM's 127-qubit kicked-Ising utility experiment on one AMD MI300X with under 0.7 GB of GPU memory and an RMS deviation of 0.032 from the hardware data
- Extended the engine beyond one GPU's memory with a hybrid CPU-storage / GPU-compute mode, and prototyped multi-GPU execution on an 8-GPU NVIDIA Blackwell node
- Ported Pauli-propagation simulation to the Tenstorrent Blackhole p150a NPU

Quantum algorithms and quantum machine learning

- Sole developer of a latent-space quantum GAN for KRAS G12D inhibitor discovery (National Research Foundation of Korea), trained on a Transformer latent space built from 1.4M augmented SMILES
- Co-first author of SAFE ma-QAOA, a three-stage method (Pauli-propagation pre-training, parameter distillation, fine-tuning) that achieved better solution quality than standard training while reducing estimated QPU workload by ~95% on average
- Co-developed mixed IQP-QCBMs, ancilla-augmented mixtures of IQP circuits for generative modeling, benchmarked from 16 to 784 qubits
- Built a neural-network error mitigation model that reduced the total variation distance to ideal outputs; studied VQE and QAOA on IBM, IonQ, and Quantinuum hardware

Sogang University, Department of Physics — Seoul, Korea
Research Assistant (Advisor: Prof. Minsoo Kim) — Sep 2024 – Dec 2024

- Analyzed quantum Hall transport in mesoscopic 2D systems (valley-dependent transport, Dirac cones)

Pohang University of Science and Technology (POSTECH), Department of Physics — Pohang, Korea
Research Intern (Advisor: Prof. Yoonho Kim) — Jul 2024

- Built a high-precision interferometer; ran Bell-test and SPDC photon-pair entanglement experiments

PUBLICATIONS & PRESENTATIONS

† Equal contribution.

Papers

(3) **H. Kim**, Y. Lee, "PADO-Pauli: A GPU-Accelerated Implementation of Pauli Propagation," (2026). arXiv:2610.17755. github.com/Norma-Q/PADO-Pauli

(2) Y. Lee, **H. Kim**, "Trainability and Mode Separation of Mixed IQP-QCBMs," (2026). arXiv:2607.27883
- Oral presentation, ML4QT Symposium 2026, *Waterloo*, Sep 23–25, 2026
- Poster, QTML 2026, *Stellenbosch*, Dec 6–11, 2026

(1) **H. Kim**†, Y. Lee†, "SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation," in *Proc. IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026 (to appear). arXiv:2605.23377
- Oral presentation, IEEE Quantum Week (QCE 2026), Technical Papers Track, *Toronto*, Sep 13–18, 2026

Posters

PADO-Pauli: ML4QT Symposium 2026 (*Waterloo*), IEEE QCE 2026 (*Toronto*) — Sep 2026
SAFE ma-QAOA: Quantum Future Korea 2026 (*Seoul*), CQI 2026 (*Seoul)* — May & Jul 2026
Generative models for drug discovery: QISK Annual Meeting; National Undergraduate Quantum Conference — Feb 2026

SELECTED PROJECTS

Quantum Reframing Challenge 2026 (MegazoneCloud), Finalist
Team of three, chemistry and materials track — 2026
- Pipeline of Pauli-propagation surrogate pretraining, QPU sampling, and sample-based quantum diagonalization (SQD) for Fe–N_2 catalysis models at 12–20 qubits, including runs on IonQ Forte

Realizing Quantum Advantage in the Generation of Drug Library by Quantum Machine Learning
National Research Foundation of Korea, Quantum Computing Challenge — 2024 – 2026
- Built the evaluation pipeline for quantum-generated molecules against KRAS G12D-binding ligands

HONORS & AWARDS

Excellence Award (2nd place among 11), KQIC Quantum Internship, National Information Society Agency — 2025
Full-Tuition Merit Scholarship, Sogang University (four years) — 2020 – 2025
Merit Scholarships:
Lotte Scholarship Foundation, Catholic Outstanding Student, Sogang Alumni Association — 2023 – 2025

TEACHING, MENTORING & OUTREACH

Technical Mentor, 1st Quantum AI Competition (NORMA & AI Factory), 55 teams — Aug 2025
Invited Talk, Waterloo Quantum Club (student organization), University of Waterloo — Jul 2026
Technical Presenter, NORMA booths at Q2B Tokyo, Quantum Korea, and IEEE Quantum Week — Jun – Sep 2026
Teaching Assistant, Introduction to AI Programming (Python), Sogang University, ~200 students — 2024 – 2025

TECHNICAL SKILLS

Programming & ML: Python (PyTorch, pandas, NumPy), C
Quantum Software: NVIDIA CUDA-Q, IBM Qiskit, pytket, PennyLane
Quantum Hardware: IBM (Eagle, Heron, Nighthawk), IonQ (Forte), Quantinuum (H2)
Hardware: NVIDIA GPUs (GB300, RTX PRO 6000, A100), AMD GPUs (MI300X, MI325X), Tenstorrent NPU (p150a)
Languages: Korean (native), English (CEFR C1, TOEIC 930)

LEADERSHIP

Sponsorship and Operations Lead
National University Jiu-Jitsu Championship (100+ athletes, 10 sponsors) — Aug 2025
Vice President, T.A.P. Brazilian Jiu-Jitsu Club, Sogang University — Mar 2025 – Jun 2025
Training Lead, T.A.P. Brazilian Jiu-Jitsu Club, Sogang University — Sep 2024 – Dec 2024

MILITARY SERVICE

Republic of Korea Army, Sniper, Military Police Special Duty Team (counterterrorism) — Feb 2021 – Aug 2022